

PROJECT LANDFALL

The Living World Navigation System

Governing Amendment v1.1

Worldspaces and Virtual Navigation

Physical Geography, Virtual Maps, Multi-Worldspace Journeys, and Unified Living Charts



Amended Governing Principle

Landfall governs place and journey, not merely GPS. A Chronicle may navigate physical geography, an authored virtual world, or a sequence of multiple Worldspaces through one canonical Living Chart model. The original zero-infrastructure physical-world rule remains fully intact: real-world Landfall must still work using only Creator tools and the Player device, while virtual navigation must never pretend that unavailable game telemetry exists.

Version 1.1 • September 21, 2026

Prepared for the Voyagewright / Chronicles platform

Document Control

Field	Value
Program	Project Landfall
Subsystem	The Living World Navigation System
Document type	Cross-cutting governing amendment to Project Landfall v1.0
Version	1.1
Date	September 21, 2026
Status	Governing amendment - final design baseline
Amends	Project Landfall Governing Architecture and Product Requirements Document v1.0, July 24, 2026
Primary new scope	Worldspaces, virtual maps, coordinate-system abstraction, virtual observation providers, multi-worldspace journeys, unified Living Chart projection
Preserved hard rule	Zero-infrastructure physical Landfall remains mandatory; hardware and installed markers stay optional
Primary adjacent projects	Storytide, Watchglass, One Voyage, Drydock, Lanternwake, Wayfarer, Creator Studio

Authority and Precedence

This amendment does not replace Project Landfall v1.0. The v1.0 document remains authoritative for real-world navigation, privacy, confidence, route authoring, offline behavior, accessibility, provider independence, progression, and the four-phase program except where this amendment explicitly broadens or supersedes a real-world-only assumption.

- When v1.0 says "map" or "location" in a way that assumes Earth geography, v1.1 generalizes the concept to a Worldspace and its coordinate/reference system.
- When v1.0 defines physical evidence, geolocation, accuracy meters, or GPS-specific behavior, those rules remain normative for PHYSICAL Worldspaces and do not automatically apply to VIRTUAL Worldspaces.
- No v1.1 rule may weaken the v1.0 zero-infrastructure requirement, privacy protections, canonical progression ownership, replay safety, accessible wayfinding, or provider-independence rules.
- If this amendment and a future accepted Landfall amendment conflict, the newer accepted amendment governs only the explicitly amended subject.

Compatibility Rule

Every valid v1.0 Landfall definition is representable in v1.1 as a PHYSICAL Worldspace without semantic loss. Migration may add explicit Worldspace identity and provider metadata, but must not reinterpret or move existing coordinates, alter waypoint completion history, or change pinned Tale Sessions.

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1. Executive Summary

Project Landfall v1.0 correctly established a rigorous real-world navigation subsystem: Creator-authored waypoints and routes, a Chronicle-native Voyage Chart, confidence-aware arrival verification, privacy-safe location handling, contextual precision, offline behavior, and canonical progression. This amendment preserves that work and broadens one foundational assumption: a meaningful Chronicle location does not always belong to Earth.

A Voyagewright Chronicle may unfold through a town, trail, museum, campus, or shoreline; it may also unfold through a virtual game world, a fictional chart, an image-backed map, or a sequence that crosses more than one realm. The Player still needs to understand where they are in the story, where they have been, where they are headed, what has been discovered, and how the journey has changed. Those are Landfall responsibilities even when latitude and longitude are meaningless.

Version 1.1 therefore introduces the Worldspace abstraction. A Worldspace is the canonical spatial reference domain within which Landfall places locations, routes, regions, observations, journey-path segments, map presentation, and navigation state. PHYSICAL Worldspaces use real geography and may consume GPS/native location. VIRTUAL Worldspaces use authored coordinates and may consume progression, Player confirmation, Captain confirmation, Watchglass, or future game integrations. A Chronicle may contain one or more Worldspaces while remaining one Tale Session and one continuous story.

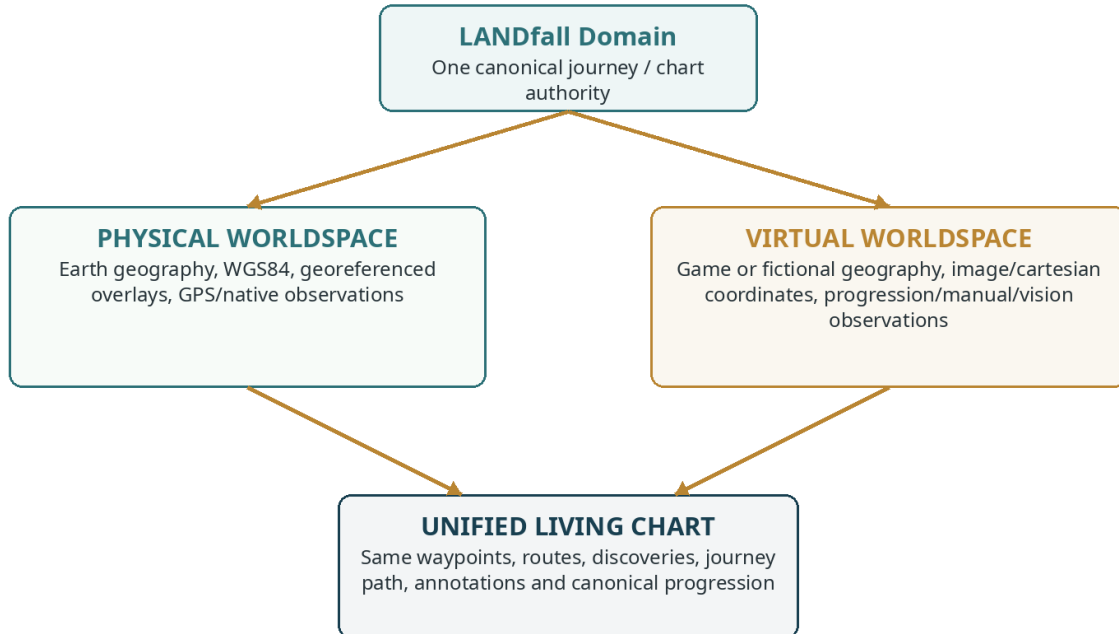


Figure 1. Worldspaces broaden Landfall from Earth-only location to a unified physical and virtual journey domain.

Core Product Outcome

A Creator can author a coherent journey in a real town or a virtual world using one Landfall model. A Player sees one Chronicle-native Living Chart showing the current objective, visited places, discoveries, route progress, annotations, and journey so far. The system knows the difference between actual observation, inferred context, manual confirmation, and unavailable evidence, and it never invents telemetry simply because a map looks convincing.

2. Amendment Intent, Scope, and Non-Goals

2.1 Intent

The amendment exists to prevent Landfall from becoming synonymous with browser geolocation. GPS is one observation provider for one class of Worldspace. It is not the definition of place. The domain must represent place first, then allow each Worldspace to declare how its places are encoded, displayed, observed, verified, and progressed.

2.2 New Governed Scope

- Worldspace identity and lifecycle inside immutable published Chronicle definitions.
- Physical and virtual spatial reference systems with explicit coordinate semantics.
- Image-backed, local-cartesian, vector, and georeferenced presentation maps.
- Multiple Worldspaces inside one Chronicle and canonical transitions between them.
- Normalized location observations from heterogeneous evidence providers.
- Virtual-map journey history, visited/discovered state, route drawing, annotations, and objective context.
- Creator authoring and preview of virtual worlds without requiring Earth coordinates.
- Storytide presentation contracts for mobile expedition and desktop immersive experiences.
- Watchglass context fusion for both physical and virtual visual verification.
- Drydock static analysis and deterministic simulation of Worldspace/provider behavior.

2.3 Explicit Non-Goals

- Landfall does not become a game telemetry reverse-engineering project.
- Landfall does not require Sea of Thieves or any other game to expose positional telemetry.
- Landfall does not scrape memory, inject into games, bypass anti-cheat, or imply access to unsupported APIs.
- Landfall does not own Storytide narrative composition, Figurehead identity rendering, Watchglass visual inference, or One Voyage progression truth.
- Landfall does not require every Chronicle to use maps or location-aware completion.

- Landfall does not require exact current-position markers in virtual worlds when only coarse or event-derived location is known.
- Landfall does not force desktop and phone experiences into identical layouts.
- Landfall does not convert a PHYSICAL Worldspace into a VIRTUAL Worldspace merely because the Creator uses fictional parchment artwork as the visible map.

3. Governing Principles and Hard Boundaries

Principle	Requirement
Place before provider	Worldspace and location semantics are canonical. Observation providers are replaceable evidence sources, not the domain model.
Physical rule preserved	Every core PHYSICAL Landfall journey still works using only Creator tools and the Player device. Optional hardware remains optional.
No invented telemetry	A VIRTUAL Worldspace must never show a precise live position because the UI would like one. Current context must reflect actual evidence.
One Living Chart	Full Voyage Chart, embedded chart, compact cues, Storytide surfaces, Captain views, replay, and Passport journey history project one Landfall domain.
One Chronicle session	Crossing Worldspaces never creates a second Tale Session, separate inventory, duplicated crew, or disconnected story history.
Coordinate honesty	Every coordinate carries an explicit reference system. Units, origin, orientation, transforms, and uncertainty are declared rather than guessed.
Observation truth	Player confirmation, Captain confirmation, Watchglass, progression inference, GPS, and future telemetry are distinct evidence classes with provenance.
Presentation adaptation	Mobile Expedition and Desktop Immersive may differ substantially in composition while preserving the same canonical journey semantics.
History is first-class	Visited places, discoveries, route segments, and Worldspace transitions may become durable safe history when the Chronicle intends it.
Privacy remains contextual	Physical coordinates receive strong minimization/privacy controls; virtual coordinates are still protected when they reveal private story state or Player behavior.
Provider independence	Rendering, map data, observation, route, transform, and optional game-integration providers remain replaceable.
Fail closed on uncertainty	Unknown Worldspace, invalid transform, missing map reference, unsupported provider, or ambiguous evidence broadens or blocks verification rather than silently completing.

Hard Boundary

Worldspace generalization must never weaken the original physical-world promise. If a town-wide Chronicle becomes harder to author, less private, less accessible, or more dependent on installed infrastructure because virtual maps were added, the amendment has failed.

4. Worldspace Domain Model

4.1 Definition

A Landfall Worldspace is an immutable published spatial domain that defines what "place" means for a portion of a Chronicle. It provides a canonical coordinate reference, map/presentation resources, privacy behavior, observation capabilities, routing semantics, and stable identity for waypoints and journey history.

```
type LandfallWorldspaceKind = "PHYSICAL" | "VIRTUAL";

type LandfallWorldspace = {
  id: string;
  taleId: string;
  name: string;
  kind: LandfallWorldspaceKind;
  coordinateReference: LandfallCoordinateReference;
  mapDefinitionIds: string[];
  defaultMapDefinitionId?: string;
  observationPolicy: LandfallObservationPolicy;
  routePolicy: LandfallRoutePolicy;
  privacyPolicy: LandfallWorldspacePrivacyPolicy;
  offlinePolicy: LandfallOfflinePolicy;
  presentationMetadata: LandfallWorldspacePresentationMetadata;
};
```

4.2 Why There Is No Canonical HYBRID Kind

A Chronicle may be hybrid, but a single canonical coordinate truth should not be vaguely half-physical and half-virtual. Hybrid experiences are represented by multiple Worldspaces or by a physical Worldspace whose visible presentation map is fictional/artistic but remains aligned to physical coordinates. This keeps observation rules, coordinate units, privacy, and verification explicit.

Concept	Correct representation
Fantasy parchment guiding a real town	PHYSICAL Worldspace + Chronicle-styled/georeferenced presentation map
Sea of Thieves island chart	VIRTUAL Worldspace + image/cartesian/vector presentation map
Real-world prologue followed by a game chapter	Two Worldspaces with a canonical Worldspace transition
Museum floor plan tied to a building	PHYSICAL Worldspace + local/georeferenced floor-map

	transform
Completely fictional dungeon map	VIRTUAL Worldspace + local coordinate reference

4.3 Identity and Versioning

- Worldspace IDs are stable inside a published Chronicle version and referenced by waypoints, routes, map definitions, observations, and history.
- Published Worldspaces are immutable and version-pinned exactly like other Chronicle definitions.
- Creator edits produce a new draft and later a new PublishedTaleVersion; active Tale Sessions remain pinned.
- Cross-version comparison may report added, removed, or transformed Worldspaces through Tideglass, but Landfall itself does not silently remap historical sessions.

5. Coordinate Systems, Reference Frames, and Transforms

5.1 Canonical Coordinate Reference Types

Reference type	Use	Required metadata
WGS84	Outdoor or global physical geography	Latitude/longitude semantics, altitude policy, provider precision rules
CUSTOM_GEOREFERENCED	Floor/site/image map aligned to physical geography	Control points or transform, source CRS, residual/error metadata
LOCAL_CARTESIAN_2D	Virtual or bounded local world	Origin, axis directions, units, bounds, rotation/orientation policy
NORMALIZED_IMAGE_2D	Image-backed virtual maps without world units	X/Y range, image dimensions/aspect, origin and axis convention
CUSTOM_VECTOR_2D	Authored vector world or game-map coordinate plane	Declared bounds, units, transform/version metadata

A future 3D or geocentric reference may be added through versioned contracts, but v1.1 does not require arbitrary 3D navigation. The schema must be extensible without allowing untyped "custom coordinates" whose meaning lives only in Creator memory.

5.2 Transform Rules

- Every presentation map that differs from canonical coordinates must declare a deterministic transform or explicitly declare that it is purely illustrative and cannot host interactive geometry.
- Round-trip transform error must be measurable where a reversible transform is claimed.
- Creator Studio must surface transform validity and alignment quality rather than hiding it behind successful image upload.

- Physical transform error is treated as location uncertainty; virtual transform error is treated as map-definition error. These are not interchangeable.
- Published snapshots bind transform version/checksum so replay and history use the same geometry the Player experienced.

5.3 Units and Distance

Landfall must distinguish metric physical distance from virtual-map distance. A virtual route may use map units, normalized distance, named travel bands, or Creator-supplied semantic distance. Player UI should not display "143 m" unless the Worldspace contract says meters are meaningful.

6. Unified Living Chart and Journey Projection

The Worldspace amendment does not create a separate "game map" subsystem. The Living Chart remains Landfall's canonical Player map surface. Its projection is generalized so the same read model can render real geography or authored virtual geography without forcing either world to impersonate the other.

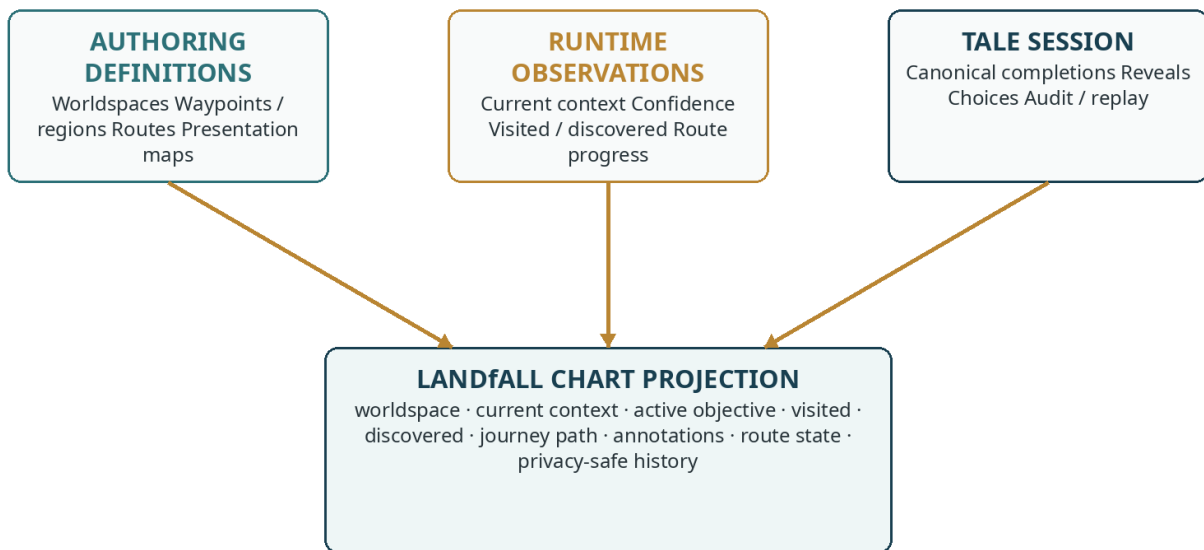


Figure 2. A Living Chart projection combines immutable definitions, current observations, and authoritative Tale Session state.

```

type LandfallChartProjection = {
  sessionId: string;
  activeWorldspaceId?: string;
  activeMapId?: string;
}
  
```

```
currentContext?: LandfallCurrentContext;
activeObjective?: LandfallObjectiveProjection;
visitedLocationIds: string[];
discoveredLocationIds: string[];
hiddenLocationCount?: number;
activeRoute?: LandfallRouteProgress;
journeyPath: LandfallJourneyPathSegment[];
annotations: LandfallChartAnnotation[];
availableWorldspaces: LandfallWorldspaceSummary[];
presentationHints: LandfallChartPresentationHints;
};
```

6.1 Current Context

Current context may be precise, coarse, inferred, or unavailable. It is not required to be a continuously moving marker. A virtual Chronicle that only knows "Crescent Isle" must show island-level context rather than inventing a dot on the north beach. A physical Chronicle with a ± 35 m GPS fix may show an accuracy region rather than a pinpoint.

6.2 Journey Path

- The journey path is a canonical Player-safe sequence of meaningful spatial transitions and confirmed locations, not necessarily a raw breadcrumb trail.
- Physical Worldspaces default to storing summarized route/progression segments, not continuous raw GPS samples.
- Virtual Worldspaces may derive path segments entirely from progression events or confirmed location observations.
- Worldspace transitions become explicit path boundaries so a hybrid Chronicle can show one continuous story without pretending coordinate systems are compatible.
- Replay and Passport/history may render the journey path from persisted safe events without reacquiring live location.

7. Normalized Waypoints, Routes, Regions, and Discoveries

The existing v1.0 waypoint and route taxonomy remains useful. Version 1.1 generalizes geometry ownership so definitions live inside a Worldspace and precision profiles become evidence profiles appropriate to that Worldspace.

v1.0 concept	v1.1 generalized interpretation
Arrival Point	Target region in any Worldspace; may be confirmed by physical or virtual evidence
Search Region	Broad target context; especially useful when exact virtual

	position is unavailable
Pass-Through Gate	Crossing/transition condition in a route plane; physical direction or virtual progression
Dwell Zone	Physical dwell or virtual/contextual sustained-presence rule when a provider supports it
Sequence Waypoint	Same semantics across Worldspaces
Optional Discovery	Same semantics; may be discovered by proximity, observation, progression, or explicit choice
Hidden Waypoint	Same semantics; hidden map state is projection policy, not absence from canonical definition
Indoor Region	PHYSICAL specialization
Natural Landmark	Usually PHYSICAL + Watchglass; virtual visual landmarks use the same Watchglass handoff with different context

7.1 Evidence Profiles

The v1.0 precision profiles remain normative for PHYSICAL Worldspaces. VIRTUAL Worldspaces use evidence profiles that describe how a location may become Possible, Nearby, Likely Inside, or Confirmed. Evidence profiles may declare accepted provider classes, minimum confidence, required corroboration, manual fallback, Captain authority, and whether exact position is meaningful.

7.2 Route Models

Ordered, flexible, branching, loop, guided-corridor, hidden, approximate, and Captain-directed routes remain canonical across Worldspaces. A route may additionally declare whether geometry is navigational, illustrative, or story-order-only. This matters when a virtual map shows a route line but Voyagewright cannot observe the Player moving continuously along it.

8. Location Observation and Provider Architecture

Version 1.1 separates spatial definition from observation. A LocationObservation is evidence that the Player is, was, or should be treated as being in a Landfall place. It always records provenance and never erases the distinction between measured, visually recognized, inferred, or human-confirmed state.

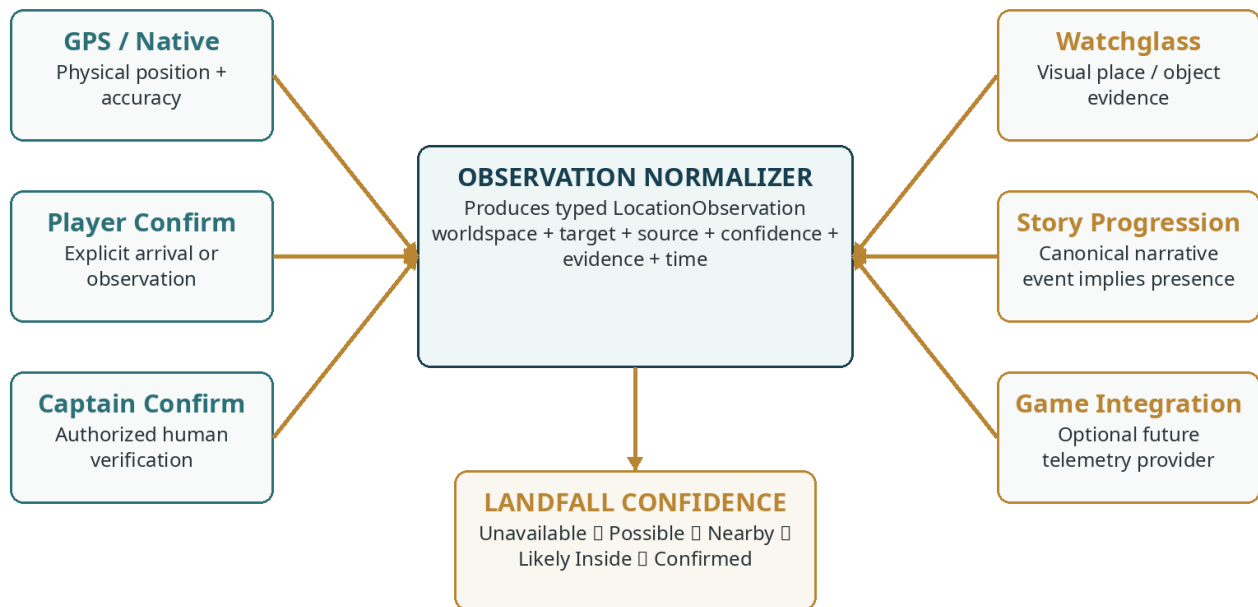


Figure 3. Heterogeneous providers normalize into typed observations before Landfall computes confidence and progression eligibility.

```

type LandfallObservationSource =
  | "BROWSER_GEOLOCATION"
  | "NATIVE_LOCATION"
  | "PLAYER_CONFIRMATION"
  | "CAPTAIN_CONFIRMATION"
  | "WATCHGLASS"
  | "STORY_PROGRESSION"
  | "CREATOR_LOGIC"
  | "GAME_INTEGRATION";

type LandfallLocationObservation = {
  id: string;
  sessionId: string;
  worldspaceId: string;
  targetLocationId?: string;
  source: LandfallObservationSource;
  observedAt: string;
  confidence: number;
  precision?: LandfallObservationPrecision;
  evidenceRef?: string;
  expiresAt?: string;
  providerVersion?: string;
};

```

8.1 Provider Rules

Provider class	Typical Worldspace	Authority and constraints
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Browser / native geolocation	Physical	Measured position; subject to accuracy, staleness, permission, spoofing and dwell rules
Player confirmation	Physical or virtual	Explicit human claim; Creator chooses whether sufficient alone or only as fallback
Captain confirmation	Physical or virtual	Authorized human verification; audit required; may satisfy configured override
Watchglass	Physical or virtual	Visual evidence; consumes Worldspace context; must preserve abstention and certification rules
Story progression	Primarily virtual	Canonical story event may establish coarse presence if the Chronicle logic guarantees it
Creator logic	Either	Typed deterministic rule; must be Drydock-validatable and not arbitrary client code
Game integration	Virtual	Optional future provider; explicit capability, provenance, versioning and failure behavior required

8.2 Observation Does Not Equal Completion

An observation changes Landfall confidence or current context. Completion still flows through LandfallProgressionService into the canonical Tale Session action path with idempotency, version pinning, visibility rules, and replay safety. A Watchglass match or Player button press is not a secret back door around One Voyage.

8.3 Observation Fusion

- Providers may corroborate or veto each other under a versioned evidence policy.
- No generic numeric average may combine incomparable evidence classes blindly.
- Physical accuracy and virtual recognition confidence remain typed separately.
- Conflicting evidence must produce uncertainty, request another provider, or require human confirmation rather than silently choosing the most convenient source.
- Provider unavailability is a normal runtime state and must be visible to Player/Captain/diagnostics where relevant.

9. Physical Worldspace Specialization

All v1.0 physical-world behavior remains normative. PHYSICAL Worldspaces use WGS84 or a georeferenced local representation and may consume browser/native geolocation, map matching,

motion, heading, elevation, Watchglass landmarks, Player observation, Captain confirmation, and optional precision providers.

- The zero-infrastructure default remains a final acceptance gate.
- Raw continuous trails remain minimized and discarded by default.
- Accuracy meters, stale-sample rules, impossible-speed rejection, hysteresis, dwell, route matching, background geofences, permission states, and physical safety guidance continue unchanged unless a future physical-specific amendment supersedes them.
- Fictional/artistic map appearance does not make the Worldspace virtual if the canonical place truth remains physical.
- Mobile Expedition is the primary presentation expectation for outdoor treasure hunts, but desktop access to safe history and story state remains supported.

10. Virtual Worldspace Specialization

A VIRTUAL Worldspace represents authored geography that is not anchored to real-world latitude/longitude: a game world, fictional island chart, tabletop map, fantasy dungeon, simulation, or other spatial narrative domain. It must be navigable and memorable without pretending Voyagewright has telemetry it does not possess.

10.1 Virtual Map Sources

- Image-backed maps using NORMALIZED_IMAGE_2D coordinates for straightforward authoring.
- Local-cartesian maps with declared bounds and units for worlds that have meaningful coordinates.
- Vector map packages for scalable labels, regions, routes, and thematic layers.
- Creator-authored overlays and Chronicle styling on top of a base virtual map.
- Future supported game-map adapters, provided licensing, attribution, security, and compatibility requirements are satisfied.

10.2 Virtual Current Position

A live current-position marker is optional. The UI must choose the most truthful representation supported by evidence: exact point, region, named location, active route segment, last confirmed location, or no current position. A named-location context such as "Crescent Isle" is preferable to a fabricated coordinate.

10.3 Virtual Journey History

- Confirmed virtual locations may be stamped/inked as visited.
- Route segments may appear after canonical travel/progression events even when continuous movement is unobserved.

- Hidden destinations remain hidden until reveal conditions succeed.
- Artifacts, Memories, notes, chapter moments, and annotations may attach to stable Landfall locations.
- Virtual coordinates may be stored more freely than real home/private coordinates, but spoiler/private-state policy still applies.

10.4 Sea of Thieves Constraint

The canonical Sea of Thieves acceptance archetype does not assume an official Player-position feed. Initial Landfall must remain useful with story progression, Player confirmation, Captain confirmation, and optional Watchglass. If a trustworthy supported game-integration provider becomes available later, it plugs into the same observation contract rather than redefining the Worldspace.

11. Multi-Worldspace and Hybrid Chronicle Journeys

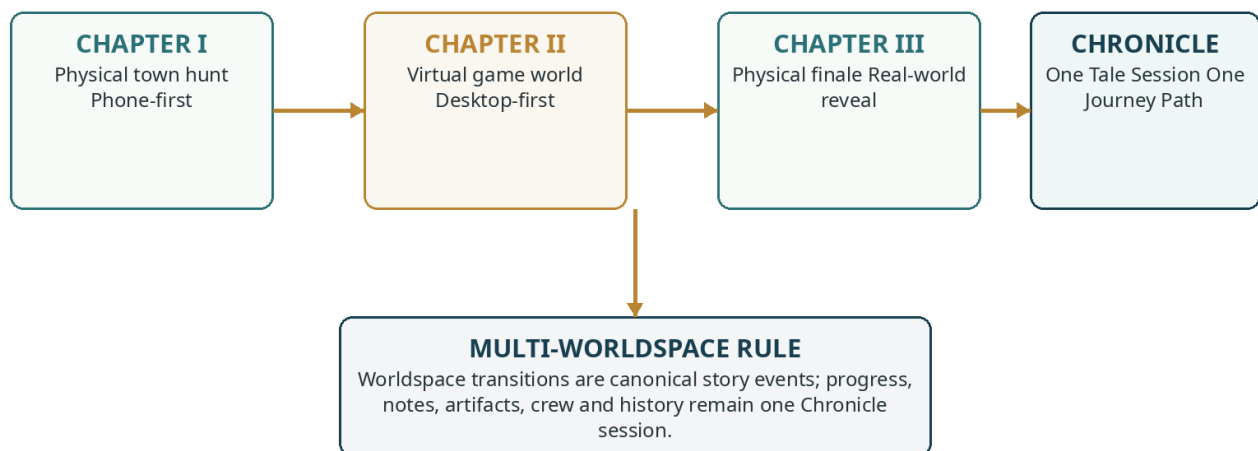


Figure 4. One Chronicle may cross multiple Worldspaces while remaining one authoritative Tale Session and one story history.

A Chronicle may reference more than one Worldspace. Worldspace transition is a canonical narrative/spatial event, not a new session. The Player keeps the same crew, variables, inventory, chapter history, notes, artifacts, and authority state.

11.1 Transition Contract

- A transition declares the source Worldspace, destination Worldspace, triggering story condition, presentation policy, and initial destination context.
- Transitions may be explicit chapter boundaries, story blocks, Captain actions, or deterministic progression consequences.
- A destination Worldspace may preload map assets before the transition but may not reveal hidden content early.
- Worldspace transition history is durable if it matters to replay or personal Chronicle history.
- Offline operation must make clear whether destination assets are available before committing a transition that would strand the Player.

11.2 Cross-Worldspace Routes

A single route definition must not pretend its geometry crosses incompatible coordinate systems. A Journey Path can contain route segments from multiple Worldspaces connected by typed transition edges. Presentation may draw them as one story timeline or separate map panels.

12. Creator Studio Authoring Experience

Creator Studio must make Worldspace choice an early, understandable authoring decision without exposing coordinate-system jargon before it is useful. The Creator should think first in terms of "real place" versus "virtual/game world" and only enter advanced reference settings when required.

12.1 New Worldspace Wizard

Step	Creator question / action
1. Realm	Is this journey happening in the real world or an authored virtual world?
2. Map source	Choose live real map, uploaded/aligned physical map, image-backed virtual map, or supported vector/custom source.
3. Coordinate/reference	Use default reference or open advanced coordinate settings.
4. Observation	How can Voyagewright know the Player reached places here?
5. Presentation	Choose Chronicle style, labels, discovery behavior, camera defaults, and Storytide hints.
6. Privacy/offline	Set classification, package behavior, retention, and public projection rules.
7. Test	Run physical test walk or virtual scenario preview before publication.

12.2 Virtual Authoring Canvas

- Upload/select a map and calibrate bounds/aspect/origin.
- Place named locations, regions, route segments, hidden places, discoveries, and annotations directly on the virtual map.
- Attach an observation policy to each location or inherit from a Worldspace default.
- Preview what the Player sees before/after each reveal.
- Preview Desktop Immersive and mobile companion/chart compositions without changing canonical geometry.
- Author route/story order separately when the visual line is illustrative rather than automatically observable.
- Run Drydock scenarios for correct destination, wrong destination, manual confirmation, Watchglass abstention, provider unavailable, replay, and Worldspace transition.

12.3 Creator Warnings

- Virtual location requires a provider that can never produce CONFIRMED.
- Exact marker is displayed even though only region-level evidence exists.
- Worldspace map has no stable coordinate reference or changed dimensions after waypoint placement.
- Route declares metric distance in a coordinate system without metric units.
- Hidden location leaks through public labels, alt text, asset filenames, or offline manifest metadata.
- Physical private geometry is being exported into a Community-safe package.

13. Player Experience and Presentation Adaptation

Landfall owns journey truth; Storytide owns how that truth is woven into the active Chronicle. The same Chart projection may be composed very differently for a walking Player holding a phone and a seated Player using a desktop beside a game.

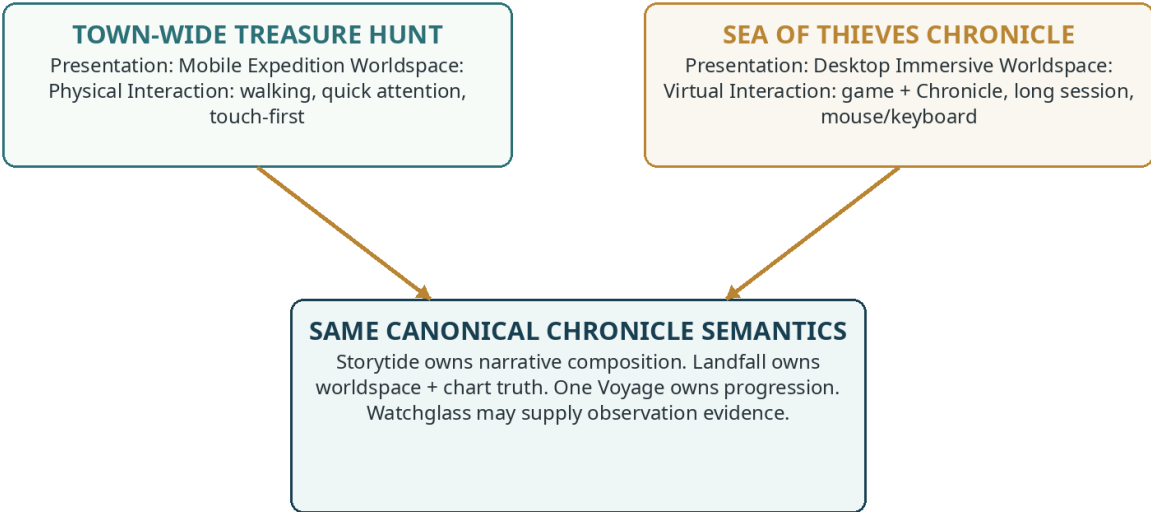


Figure 5. Two canonical Storytide presentation profiles consume the same Landfall semantics without pretending the interaction posture is identical.

13.1 Shared Requirements

- The active objective and spatial context are never hidden behind an unexplained tab hierarchy.
- Opening the full Chart must preserve story position and allow a clear return to the current narrative beat.
- Visited/discovered/hidden states use consistent semantics across phone, desktop, replay, Captain, and Passport views.
- Hover may enhance desktop but must never carry required information unavailable to touch/keyboard users.
- Cross-device access reads the same canonical session state; device recommendation does not create separate progress.

14. Mobile Expedition Requirements

For outdoor/town-wide Chronicles, phone is a primary execution environment rather than a responsive afterthought. Landfall must expose low-attention, thumb-first, sunlight-tolerant navigation that coexists with Storytide narrative presentation.

Concern	Requirement
Attention	Current objective and next spatial action readable within seconds; avoid forcing long map inspection while walking.

Touch	Primary controls have comfortable targets and no hover dependency.
Chart access	One-action access to current objective/route; easy return to Storytide narrative.
Permissions	Contextual location request with plain-language reason and visible pause/fallback controls.
Battery	Suspend expensive rendering/tracking when unnecessary; expose useful degradation rather than silently draining power.
Connectivity	Downloaded story/map content and queued evidence support weak/offline operation.
Motion	Shorter/economical map camera motion; reduced-motion remains complete and understandable.
Environment	High contrast, readable type, bright-light resilience, one-handed use, orientation changes, safe walking prompts.

14.1 Mobile Composition Guidance

- Compact objective card may remain persistently reachable.
- Chart often behaves as a full-screen or bottom-sheet transition rather than a permanent side-by-side panel.
- Quick Player notes may use a bottom sheet or short capture flow.
- Artifact inspection prioritizes one focused object/action at a time.
- Crew/presence information may live in a compact drawer rather than compete with navigation.

15. Desktop Immersive Requirements

For a game-adjacent Chronicle, desktop/laptop is often the primary execution posture. Landfall should support a large cinematic Chart, quick context recovery after alt-tab, optional compact companion layouts, and efficient keyboard/mouse interaction without assuming the Player wants a conventional GIS interface.

- Large Chart compositions may show journey history, annotations, story landmarks, and discovered routes simultaneously.
- Compact "return to game" or side-panel patterns may be offered by Storytide while preserving the same Landfall projection.
- Keyboard shortcuts may center current objective, open/close Chart, cycle relevant locations, or return to story when those shortcuts do not conflict with accessibility/browser behavior.
- Hover details are enhancements only and must have focus/click equivalents.
- If current virtual position is coarse, the desktop map must still present named-area context honestly instead of fabricating a moving Player icon.
- Long sessions should avoid unnecessary continuous animation when the Chart is not visible.

16. Storytide Integration Contract

Storytide is expected to become the Living Chronicle Experience. Landfall provides spatial truth and navigation projections; Storytide decides how story, chapters, Journal, discoveries, notes, artifacts, crew, and Chart composition flow around that truth.

Boundary	Landfall owns	Storytide owns
Current objective location	Typed spatial target / route / context	How objective is introduced, narrated and composed
Chart state	Visited, discovered, hidden, route progress, journey path	Transition between story and Chart, visual narrative framing
Arrival	Verified spatial outcome proposal + progression request	Arrival ceremony and narrative consequence after canonical success
Device posture	Presentation hints/capabilities	Mobile Expedition / Desktop Immersive composition
Annotations	Spatial anchors and safe metadata	Narrative presentation, note-writing experience, discovery ceremony
Worldspace transition	Canonical source/destination spatial event	Cinematic/story transition between realities

Integration Rule

Storytide may make Landfall feel magical. It may not make Landfall less truthful. A beautiful cinematic cannot silently upgrade "possibly on Crescent Isle" into "exactly at the northern arch" without evidence.

17. Watchglass Integration Contract

Watchglass provides visual evidence; Landfall provides spatial context. The integration should improve both systems without creating circular certainty. Landfall may narrow what Watchglass should look for based on current Worldspace/region. Watchglass may provide evidence that refines Landfall context. Neither may silently treat the other's prior as independent confirmation.

- Watchglass observations reference the exact Worldspace and target/location version.
- Landfall context may select a Watchglass model/package but does not count as visual evidence itself.
- Watchglass confidence, abstention, compatibility, source-frame policy, and certification remain owned by Watchglass.
- Physical use: confirm an existing landmark, exhibit, room feature, or object after Landfall establishes surrounding context.
- Virtual use: recognize a game location/landmark/state from the screen, where allowed and technically supported.

- If Watchglass is unavailable, the Chronicle must have the configured alternate path: Player confirmation, Captain confirmation, progression, broader region, or another provider.

18. One Voyage Progression and Completion Semantics

One Voyage remains authoritative for live Chronicle progression. Landfall can establish spatial evidence and request a completion action; it cannot directly mutate chapter/block availability or inventory.

```
type LandfallCompletionProviderV2 = {  
  provider: "landfall";  
  worldspaceId: string;  
  locationId: string;  
  requiredOutcome: "NEARBY" | "LIKELY_INSIDE" | "CONFIRMED";  
  acceptedObservationSources?: LandfallObservationSource[];  
  minimumCorroboration?: number;  
  dwellSeconds?: number;  
  allowCaptainOverride: boolean;  
  allowPlayerFallback: boolean;  
  replayPolicy: "PRESENTATION_ONLY" | "NONE";  
};
```

- Completion requests include expected session revision/idempotency identity and pinned published-definition identity.
- A Worldspace transition caused by completion is recorded through canonical event semantics, not client-local map state.
- Replay never reacquires location or observation evidence.
- Captain correction/override remains governed, audited, and distinct from evidence source history.
- Player-visible map state must follow canonical reveal/completion state, not optimistic local action.

19. Drydock Validation and Simulation

Drydock must understand Landfall v1.1 contracts so virtual maps do not become a validation blind spot. Worldspace definitions, transforms, provider capability, route semantics, hidden-state rules, and Story Block completion policies require static and scenario validation.

19.1 Static Rules

- Every waypoint/route/map references an existing Worldspace.

- Coordinate/reference values validate against declared Worldspace kind and bounds.
- Presentation map transforms are valid and checksummed.
- Virtual route units are not mislabeled as physical meters.
- Completion provider requires an observation path capable of the requested outcome or a governed fallback.
- Hidden locations do not leak through accessible names, public projections, map labels, offline manifests, or package metadata.
- Worldspace transitions cannot create unreachable required branches or orphan the current objective.

19.2 Simulation Scenarios

- Physical: permission denied, weak accuracy, stale fix, route deviation, offline evidence, Captain override.
- Virtual: Player confirms correct/wrong place, Watchglass confirms/abstains, progression infers coarse context, provider unavailable, manual fallback.
- Multi-worldspace: successful transition, destination package unavailable, replay transition, reconnect across transition boundary.
- Cross-project: Watchglass context package mismatch, Storytide presentation replay, One Voyage duplicate completion, history reconstruction.

20. Offline, Weak-Network, and Local-Map Operation

Virtual Worldspaces make offline support easier in some cases because the entire map may be a bounded Chronicle asset. That convenience must be designed, not assumed.

- Image/vector virtual map assets may be included in authorized Chronicle offline packages with version/checksum identity.
- Physical maps continue to use region manifests/cache policy and must respect provider licensing and attribution.
- Local observations may update temporary context offline; canonical completion remains clearly queued until authoritative reconciliation when required.
- Worldspace transitions should warn before entering a destination whose required offline assets are unavailable.
- Offline history must distinguish locally observed state from server-confirmed progression.
- Reconnect reconciles idempotently without duplicating visits, discoveries, route segments, or completion events.

21. Privacy, Security, and Data Minimization

Data class	Default treatment
Physical live samples	Process locally where possible; short-lived; do not persist continuous trail by default
Physical private waypoint geometry	Protected; generalized or excluded from public/community projections
Virtual coordinates	May be retained as Chronicle state when needed, but hidden/spoiler/private locations remain protected
Observation evidence	Store minimal bounded receipt/provenance; provider-specific raw evidence follows owner project policy
Watchglass frames	Governed by Watchglass retention/consent; Landfall stores references/outcomes, not unrestricted copies
Captain overrides	Audited with actor, reason/policy, target and canonical result
Journey history	Persist semantic visited/discovery/path events; avoid raw physical breadcrumb history unless explicitly authorized

- Virtual maps must not accidentally embed private Creator notes, hidden solutions, account IDs, local file paths, or unrevealed targets in client-accessible metadata.
- Optional game integrations must have explicit authentication, least privilege, rate/security policy, and no anti-cheat bypass.
- Worldspace IDs and map asset URLs must be authorization-checked; knowledge of an ID is not access.
- Public/community packages receive a privacy/spoiler-safe Landfall projection, not the full Creator definition by default.

22. Accessibility and Inclusive Navigation

Worldspace generalization increases the need for accessible parallel semantics. A virtual map can be just as inaccessible as a real map if the experience reduces the story to unlabeled icons and animated parchment.

- Every Living Chart has a structured text representation of current context, objective, known locations, visited locations, route/progress state, and available actions.
- Hidden/unrevealed locations remain omitted or safely generalized in the accessible structure.
- Direction/distance language adapts to the Worldspace. Virtual maps without metric units use named locations, route steps, relative direction, or Creator-authored language.
- Keyboard/focus can reach all map-adjacent controls and visible interactive locations.

- Color, motion, particle effects, route drawing, marker movement, sound, or vibration are never the sole carriers of state.
- Reduced-motion mode uses staged cuts/highlights rather than long fly-throughs and remains semantically complete.
- Mobile Expedition must support screen readers, text zoom, high contrast/forced colors where applicable, and large touch targets while preserving safe walking use.

23. Animation and Experience Language

The v1.0 Lanternwake ownership boundary remains. The renderer owns map-space camera and symbol transitions; Lanternwake/Storytide coordinate narrative transitions. Version 1.1 adds Worldspace-aware scene semantics without creating a second animation stack.

- `landfall-worldspace-enter` - arrival into a newly active Worldspace after canonical state changes.
- `landfall-worldspace-transition` - transition between Worldspaces; Storytide may own the full cinematic while Landfall provides source/destination state.
- `landfall-virtual-location-confirmed` - virtual-place stamp/reveal after authoritative success.
- `landfall-journey-path-reveal` - reveals a new historical route segment after canonical event state permits it.
- Existing physical scenes such as map arrival, location acquiring, nearby, route reveal, deviation, offline, floor-map transition, and natural-landmark confirmation remain valid.

Operational language must remain truthful. Physical contexts may say "The chart is finding your position." Virtual contexts might say "Current island not confirmed," "Crescent Isle marked from your progress," or "Confirm when your crew arrives." Do not say "GPS unavailable" inside a game world unless somebody has achieved truly remarkable cartographic confusion.

24. Canonical Data Model and Service Contracts

```
type LandfallCoordinateReference =
  | { type: "WGS84" }
  | { type: "CUSTOM_GEOREFERENCED"; transformId: string }
  | { type: "LOCAL_CARTESIAN_2D"; unit: string; bounds: Bounds2D; axis: AxisConvention }
  | { type: "NORMALIZED_IMAGE_2D"; imageAssetId: string; axis: AxisConvention }
  | { type: "CUSTOM_VECTOR_2D"; referenceId: string; bounds: Bounds2D };

type LandfallMapDefinition = {
  id: string;
  worldspaceId: string;
  role: "PRIMARY" | "FLOOR" | "DETAIL" | "OVERVIEW" | "ILLUSTRATIVE";
```

```

coordinateReference: LandfallCoordinateReference;
rendererPolicy: LandfallMapRendererPolicy;
style: LandfallStyleDefinition;
attribution?: LandfallAttribution[];
transform?: LandfallMapTransform;
offlinePolicy: LandfallOfflinePolicy;
};

```

```

type LandfallLocation = {
  id: string;
  worldspaceId: string;
  name: string;
  geometry: LandfallWorldGeometry;
  locationType: LandfallWaypointType;
  evidenceProfile: LandfallEvidenceProfile;
  visibilityPolicy: LandfallVisibilityPolicy;
  sequencePolicy: LandfallSequencePolicy;
  completionPolicy: LandfallCompletionPolicy;
  fallbackPolicy: LandfallFallbackPolicy;
  privacyClassification: LandfallLocationClassification;
};

```

```

type LandfallRuntimeStateV2 = {
  sessionId: string;
  activeWorldspaceId?: string;
  activeMapId?: string;
  activeLocationId?: string;
  activeRouteId?: string;
  currentContext?: LandfallCurrentContext;
  confidence: LandfallConfidence;
  observationState: LandfallObservationState;
  routeProgress?: LandfallRouteProgress;
  visited: LandfallVisitState[];
  discovered: LandfallDiscoveryState[];
  journeyPath: LandfallJourneyPathSegment[];
  offlineState: LandfallOfflineState;
};

```

24.1 Service Boundary Amendments

Service	v1.1 responsibility
LandfallDefinitionService	Worldspace/map/location/route authoring, validation, versioning, snapshot construction
LandfallMapService	Renderer/data selection, transforms, style compilation, map asset/attribution policy
LandfallObservationEngine	Provider intake, normalization, provenance, expiry, evidence fusion, current context
LandfallLocationEngine	Physical geometry, route matching, confidence/dwell;

	physical specialization of observation/context rules
LandfallProgressionService	Worldspace-aware verified outcomes → canonical One Voyage actions
LandfallProjectionService	Storytide/Player/Captain/Creator/replay/Passport-safe Chart projections
LandfallOfflineService	Physical regions + virtual map assets + transition readiness + reconciliation
LandfallPrivacyPolicy	Physical coordinate minimization plus virtual spoiler/private-state classification
LandfallTelemetryService	Sanitized provider/context diagnostics with Worldspace provenance

25. API and Action Surface

Action	Purpose / amendment
GET chart projection	Worldspace-aware current Chart state; one API shape for physical and virtual presentation
GET worldspace definition/projection	Authorized published definition with role-safe hidden/private filtering
POST observation	Submit bounded typed observation/receipt from an allowed provider
POST location completion	Request canonical completion using worldspace/location identity and idempotency
POST Captain override	Worldspace-aware confirm/skip/correct/replace where policy permits
POST Player confirmation	Explicit manual confirmation with exact target/policy and anti-duplicate identity
POST Creator test receipt	Physical test walk or virtual scenario evidence; protected Creator-only data
GET offline package manifest	Physical map regions and/or virtual map assets for pinned published version
POST privacy consent	Optional live physical sharing / bounded diagnostics; does not grant unrelated private access
POST worldspace transition readiness	Optional preflight for destination assets/provider capability before committing story transition

26. Events, Replay, History, and Journey Path

Existing v1.0 durable events remain valid. Version 1.1 adds only events necessary to preserve spatial narrative truth across Worldspaces.

- `LANDFALL\_WORLDSPACE\_ENTERED` - canonical active spatial realm changed.
- `LANDFALL\_WORLDSPACE\_LEFT` - optional paired event when needed for history/telemetry; avoid redundant noise.

- `LANDFALL\_LOCATION\_OBSERVED` - durable only when an observation materially affects Player-visible state or audit; ephemeral observations stay runtime-only.
- `LANDFALL\_LOCATION\_CONFIRMED` - generalized form of waypoint confirmation; legacy event maps/upcasts without loss.
- `LANDFALL\_JOURNEY\_PATH\_SEGMENT\_ADDED` - optional durable summarized travel/history segment.
- `LANDFALL\_DISCOVERY\_REVEALED` - when a spatial discovery becomes canonical Player-visible state.

26.1 Replay

Replay uses stored safe event/projection data. It never asks GPS again, reruns Watchglass, re-prompts a Player to confirm, contacts a game integration, or changes progress. A virtual route can be redrawn from history exactly as a physical arrival can be restamped from history.

26.2 Personal History

Wakebook/Chronicle Passport may consume safe Journey Path and location history as a personal archive. Landfall supplies source-bound spatial facts; Wakebook owns emotional/history presentation. Private real-world coordinates must remain generalized according to Landfall policy.

27. Migration and Historical Compatibility

27.1 v1.0 → v1.1 Mapping

v1.0 field/concept	v1.1 treatment
LandfallMap.taleId	Map belongs to explicit PHYSICAL Worldspace belonging to Tale
coordinateSystem=WGS84	Worldspace/reference remains WGS84
coordinateSystem=CUSTOM_GEOREFERENCED	PHYSICAL Worldspace + explicit transform/reference
LandfallWaypoint.mapId	Location references Worldspace and compatible map projection
LocationProvider	Becomes a PHYSICAL observation provider class
Current live fix	CurrentContext/observation evidence inside PHYSICAL Worldspace
Waypoint completion event	Upcast/alias to generalized location confirmation while preserving historical identity
Voyage Chart projection	Extends with worldspaceId and generalized current context; same canonical surface

27.2 Historical Safety

- No migration changes geometry numerically unless an explicit tested coordinate transform is part of the migration.
- No active session is silently moved to a new Worldspace definition.
- Historical event IDs remain readable and reconstruct the same Player-visible state.
- Replay of a v1.0 Chronicle does not require virtual provider code.
- Community releases and private packages identify schema/version support and fail closed when unsupported.

28. Testing and Acceptance Matrix

28.1 Core Domain Tests

- Worldspace identity/versioning and physical/virtual kind validation.
- Coordinate bounds, axis conventions, unit handling, image normalization, georeferenced transforms, transform error.
- Waypoint/route geometry validity in each reference type.
- Observation source validation, expiry, provenance, confidence, corroboration and conflicting evidence.
- Journey Path construction, Worldspace transition edges, replay reconstruction and idempotency.
- Hidden/spoiler/private projections and public-package sanitization.

28.2 Creator Tests

- Create PHYSICAL and VIRTUAL Worldspaces through Guided and advanced flows.
- Upload an image-backed virtual map, define origin/axes, place and move locations, resize regions, draw routes, hide/reveal locations.
- Align a physical custom map without accidentally converting the Worldspace kind.
- Change map artwork while preserving or intentionally migrating canonical geometry.
- Receive warnings for impossible provider/outcome combinations and unobservable exact virtual targets.
- Preview mobile/desktop Chart presentation and run deterministic virtual scenarios.

28.3 Player Matrix

Dimension	Required cases
Worldspace	Physical, virtual image-backed, virtual local-cartesian, multi-worldspace transition
Observation	GPS, denied GPS, Player confirm, Captain confirm, Watchglass confirmed/abstained, progression inferred, provider unavailable
Current context	Exact, region-only, named-location-only, last-known, unavailable

Journey	Ordered, flexible, branch, optional discovery, hidden destination, route deviation, cross-worldspace
Network	Online, slow, offline-ready, offline-partial, reconnect, duplicate evidence
Presentation	Mobile Expedition, Desktop Immersive, tablet/general responsive, phone companion read-only
Input/accessibility	Touch, mouse, keyboard, screen reader, reduced motion, high contrast, 200% zoom
Replay/history	Arrival replay, transition replay, journey path, Passport-safe projection

28.4 Security/Privacy Tests

- Physical private coordinates never leak into virtual/public projections through generic Workspace serialization.
- Hidden virtual coordinates/labels do not leak before reveal.
- Provider evidence cannot be submitted for another session/worldspace/target.
- Game-integration adapters cannot select arbitrary upstream hosts or execute untrusted Creator code.
- Watchglass evidence references correct certified package and Workspace target.
- Captain/Player manual confirmation respects capability, membership, scope and idempotency.

29. Implementation-Phase Reconciliation

The original four Landfall phases remain the program structure. Version 1.1 broadens their acceptance scope rather than inventing a parallel five-phase virtual-map project.

Phase	v1.1 amendment
Phase 1 - Lay the Bearings	Workspace domain, coordinate references, generalized map definition, observation-provider seam, physical v1.0 migration compatibility, one basic virtual image-backed map projection.
Phase 2 - Bring the World Aboard	Creator virtual-map authoring, Living Chart across physical/virtual, routes/discoveries, Player/Captain confirmation, Storytide presentation hints, multi-worldspace transition foundation, offline virtual assets.
Phase 3 - Read the Ground	Watchglass/Context fusion generalized to physical and virtual; richer evidence policies; natural/visual landmarks; exact-target honesty; game-screen visual verification where Watchglass permits.
Phase 4 - Hold the Bearing	Native physical providers remain primary scope; production provider operations also cover virtual packages, optional game integrations, cross-worldspace offline behavior, monitoring, privacy review and final dual-archetype acceptance.

Mainline Safety Requirement

Each Landfall phase remains independently coherent if the next phase never happens. Virtual-map support may land additively, but a phase may not leave existing physical Chronicles unable to publish or play merely because a future virtual feature is incomplete.

30. Canonical Experience Archetypes

30.1 Archetype A - Town-Wide Living Treasure Hunt

Dimension	Acceptance archetype
Primary device	Phone / portrait mobile-first
Worldspace	PHYSICAL
Map truth	Real geography with Chronicle styling
Observation	Foreground GPS/native location + confidence/dwell; optional Watchglass/manual/Captain
Interaction posture	Walking, intermittent attention, one-handed touch, outdoor conditions
Storytide expectation	Story remains immersive while physically moving; Chart is frequent and quick, not a detached GIS app
Failure expectation	Permission denied/weak accuracy/offline never strand the Chronicle

30.2 Archetype B - Sea of Thieves Chronicle

Dimension	Acceptance archetype
Primary device	Desktop/laptop-first; second monitor/alt-tab friendly
Worldspace	VIRTUAL
Map truth	Authored Sea of Thieves game-world chart/image/vector map
Observation	Story progression, Player confirmation, Captain confirmation, optional Watchglass; no assumed live telemetry
Interaction posture	Long seated session alongside game, mouse/keyboard, rich Chart inspection
Storytide expectation	Voyagewright feels like a narrative layer around the game rather than an unrelated website repeatedly consulted
History expectation	Visited islands/locations, discoveries, artifacts/notes and route-so-far form a meaningful virtual journey record

30.3 Cross-Archetype Acceptance

A Landfall architecture that works beautifully for the town hunt but requires special one-off code for the virtual Chronicle is incomplete. The reverse is equally unacceptable. Both archetypes must consume the same Worldspace, map, location, observation, Chart, route, progression, history, privacy, and replay contracts with specialized providers and presentation policies.

31. Final Acceptance Criteria

Gate	Required result
v1.0 preservation	Every existing physical Landfall requirement remains supported; zero-infrastructure physical Chronicle still passes final acceptance.
Worldspace truth	Every Landfall map/location/route belongs to an explicit versioned Worldspace with a declared coordinate reference.
Virtual usefulness	A complete virtual Chronicle journey can be authored and played without GPS or game telemetry.
No invented position	UI never presents more virtual/physical precision than observation evidence supports.
One Living Chart	Physical, virtual, embedded, full, Storytide, Captain, replay and history views project one canonical Landfall domain.
Creator authoring	Creators can visually author image-backed virtual locations/routes and physical maps without hand-editing coordinates.
Observation providers	GPS/native/manual/Captain/Watchglass/progression/future game adapters use one typed evidence boundary with provenance.
Multi-worldspace	One Chronicle may transition between Worldspaces without splitting Tale Session, crew, notes, inventory or history.
Mobile Expedition	Town-wide phone-first archetype passes usability, permissions, offline, accessibility, battery/performance and progression tests.
Desktop Immersive	Sea of Thieves desktop archetype passes virtual Chart, journey history, game-adjacent interaction and no-telemetry fallback tests.
Storytide contract	Narrative presentation may differ by device posture without duplicating spatial truth.
Watchglass contract	Visual evidence can refine Landfall in physical/virtual worlds without circular confidence or loss of abstention.
Drydock	Static validation and deterministic scenarios cover Worldspace/reference/provider/transition contracts.
Privacy	Physical coordinate minimization is preserved; virtual hidden/spoiler state is not leaked.
Offline/replay	Downloaded virtual maps and physical regions behave correctly; replay never reacquires evidence.
Historical compatibility	v1.0 sessions/events/replays remain readable and semantically stable.

Amended Completion Rule

Landfall is not complete because GPS works, nor because a pretty fictional map can be displayed. It is complete when Voyagewright can truthfully guide and remember a journey wherever that journey exists - physical or virtual - through one understandable, accessible, privacy-safe, provider-independent spatial system.

32. Recommended Technical Baseline

Area	Recommended v1.1 baseline
Physical renderer	MapLibre GL JS / MapLibre Native or equivalent structured vector renderer, preserving v1.0 provider independence
Virtual image maps	Standard image assets with normalized coordinate reference, bounded transforms, responsive high-resolution rendering and Chronicle overlays
Virtual vector maps	GeoJSON-like or equivalent typed 2D feature model in local/custom coordinates; do not require geographic CRS
Transforms	Deterministic versioned transform service/library with validation, residual/error reporting and round-trip tests where applicable
Geometry	Typed geometry utilities parameterized by reference/units; avoid assuming every point is latitude/longitude
Observation	Centralized provider registry + normalizer + evidence policy; PHYSICAL location engine remains specialized provider consumer
State	Canonical server projections + bounded client runtime; no map-component truth
Offline	Versioned manifest capable of map regions and self-contained virtual map packages
Storytide handoff	Typed Chart/current-context projection and presentation hints; no direct Storytide dependency inside core geometry
Watchglass handoff	Typed contextual target + evidence receipt references; raw frame ownership remains Watchglass
Drydock	Schemas/rules/scenarios registered as governed extension contracts

32.1 Technical Caution

Do not force all virtual maps through geospatial libraries that assume longitude wraps, meters, Earth curvature, or geographic projections. Reuse generic geometry algorithms where correct, but make coordinate semantics explicit. Conversely, do not reimplement physical geodesy as generic pixel math merely to share code.

33. Glossary

Term	Meaning
Worldspace	Versioned canonical spatial reference domain in which Landfall locations, routes, maps and observations have meaning.
PHYSICAL Worldspace	Worldspace whose canonical place truth is real geography or a physically aligned local reference.
VIRTUAL Worldspace	Authored non-Earth spatial domain such as a game/fantasy/image-backed world.
Hybrid Chronicle	Chronicle that uses multiple Worldspaces or a physical Worldspace with fictional presentation; not a vague coordinate type.
Coordinate Reference	Typed definition of origin, axes, units, bounds and/or physical CRS for a Worldspace/map.
Presentation Map	Renderable visual representation of a Worldspace; may differ from canonical coordinate reference through an explicit transform.
Location Observation	Typed evidence that contributes to Landfall current context or target confidence.
Observation Provider	Source of evidence such as GPS, Player, Captain, Watchglass, progression or future game integration.
Current Context	Best truthful spatial state Landfall can presently claim; may be exact, region-level, named-location, last-known or unavailable.
Journey Path	Privacy-safe semantic sequence of meaningful spatial visits/routes/transitions, not necessarily a continuous trail.
Mobile Expedition	Storytide phone-first presentation posture for physically moving Players.
Desktop Immersive	Storytide desktop/laptop presentation posture for long seated/game-adjacent play.
Worldspace Transition	Canonical change from one active spatial realm to another within the same Tale Session.

34. References and Normative Closure

1. Project Landfall - The Living World Navigation System, Governing Architecture and Product Requirements Document v1.0, July 24, 2026. Primary baseline amended by this document.
2. Project Drydock - The Chronicle Verification and Simulation System. Governs static validation, deterministic scenarios, and publishing evidence consumed by Landfall authoring contracts.
3. Project Watchglass governing documents. Governs vision evidence, safe abstention, package certification, runtime visual verification, and its own privacy/retention boundaries.

4. Project Lanternwake completion/governing records. Governs animation runtime truth and reduced-motion authority used by Landfall scenes.
5. Project One Voyage canonical progression records. Governs Tale Session events, idempotency, progression authority, replay and visibility.
6. Project Wayfarer/Wakebook records. Govern identity and durable personal history presentation consuming safe Landfall journey facts.
7. Owner-approved Landfall Worldspace design discussion, September 21, 2026: town-wide mobile physical Chronicle and Sea of Thieves desktop virtual Chronicle as canonical acceptance archetypes.

Final Governing Rule

Landfall must model the journey before it models the sensor. The real world, a game world, and a fictional chart may use different coordinate systems and evidence providers, but Voyagewright should still understand them through one coherent language of Worldspaces, places, routes, discoveries, observations, and journey history. The map is not the truth. The governed journey is.